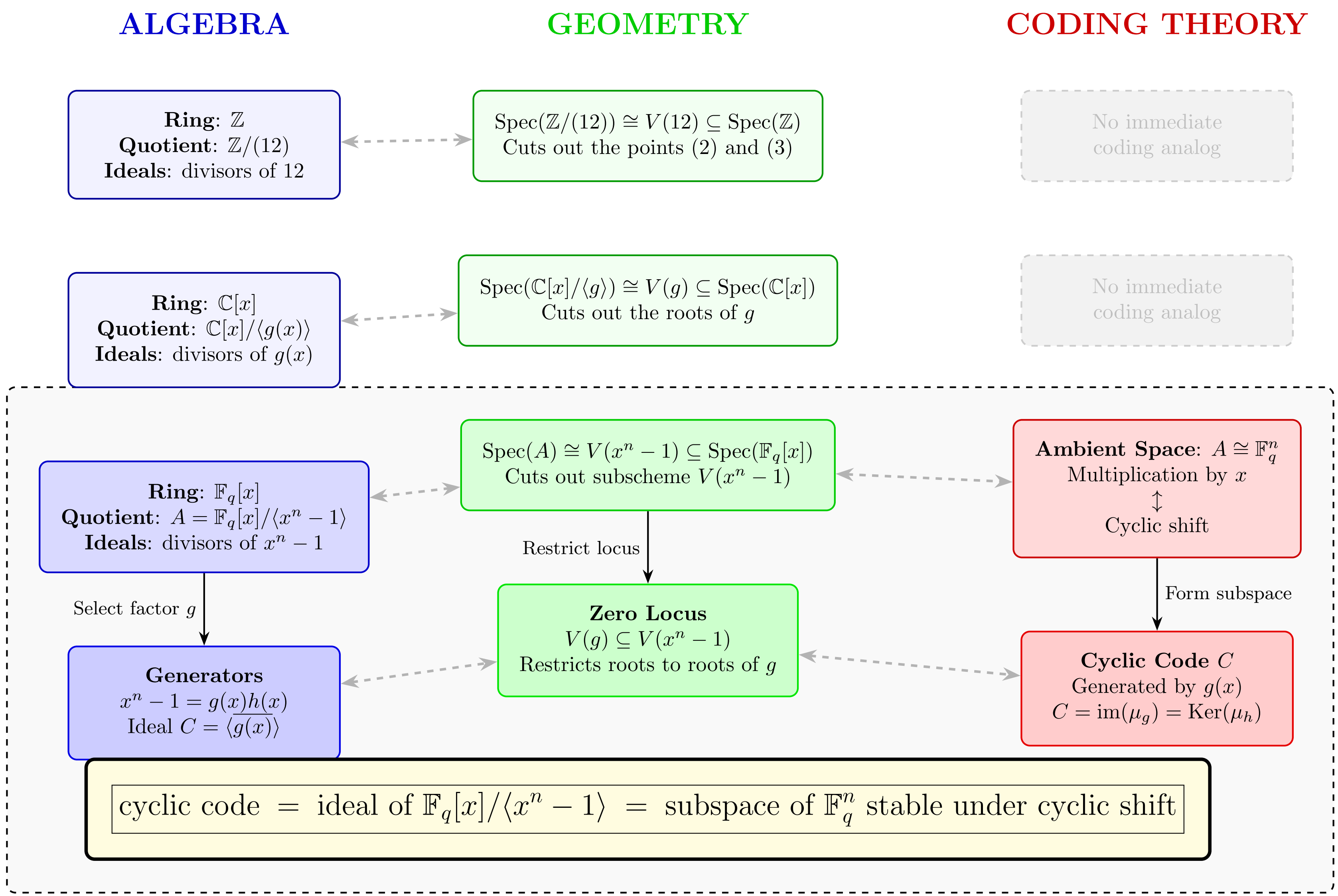
ALGEBRA
GEOMETRY
CODING THEORY
Ring: $\mathbb{Z}$
Quotient: $\mathbb{Z}/(12)$
Ideals: divisors of 12
$\mathrm{Spec}(\mathbb{Z}/(12)) \cong V(12) \subseteq \mathrm{Spec}(\mathbb{Z})$
Cuts out the points (2) and (3)
No immediate coding analog
Ring: $\mathbb{C}[x]$
Quotient: $\mathbb{C}[x]/\langle g(x)\rangle$
Ideals: divisors of $g(x)$
$\mathrm{Spec}(\mathbb{C}[x]/\langle g\rangle) \cong V(g) \subseteq \mathrm{Spec}(\mathbb{C}[x])$
Cuts out the roots of g
No immediate coding analog
Ring: $\mathbb{F}_q[x]$
Quotient: $A = \mathbb{F}_q[x]/\langle x^n - 1\rangle$
Ideals: divisors of $x^n - 1$
$\mathrm{Spec}(A) \cong V(x^n - 1) \subseteq \mathrm{Spec}(\mathbb{F}_q[x])$
Cuts out subscheme $V(x^n - 1)$
Ambient Space: $A \cong \mathbb{F}_q^n$
Multiplication by x
$\updownarrow$
Cyclic shift
Select factor g
Restrict locus
Form subspace
Generators
$x^n - 1 = g(x)h(x)$
Ideal $C = \langle g(x)\rangle$
Zero Locus
$V(g) \subseteq V(x^n - 1)$
Restricts roots to roots of g
Cyclic Code C
Generated by $g(x)$
$C = \mathrm{im}(\mu_g) = \mathrm{Ker}(\mu_h)$
cyclic code = ideal of $\mathbb{F}_q[x]/\langle x^n - 1\rangle$ = subspace of $\mathbb{F}_q^n$ stable under cyclic shift